

Lesson 1: Foundations of Logic and Proofs

Discrete Structures & Data Types

- **Discrete Structure:** Math that deals with **separate, distinct parts** rather than smooth, continuous ranges.
- **Discrete Data:** Information you **count** in exact, separate steps or whole numbers (e.g., number of students, coins, or chairs).
- **Continuous Data:** Information you **measure** along an unbroken scale, often using decimals (e.g., height, weight, time).

Logic, Proofs, and Theorems

- **Logic:** The strict rulebook for correct thinking and reasoning, used to build software and digital hardware.
- **Proof:** A step-by-step logical chain of statements that definitively shows whether a claim is true or false.
- **Theorem:** A mathematical statement that has been proven true beyond any doubt.

Propositions

A **proposition** is any statement that is strictly **True or False**—it can never be both or in-between.

- **Valid Propositions:**
 - "The moon is made up of white cotton." (False)
 - "Manila is the capital of China." (False)
 - " $1 + 0 = 1$ " (True)
- **Not Propositions:**
 - Questions: "*Bossing, is that you?*"
 - Open sentences with unspecified variables: $X + 1 = 2$
 - Commands / Imperatives: "*Read this carefully.*"

Arguments & Validity

- An **argument** combines given setup statements (premises) to see if they force a specific takeaway (conclusion) to be true.
 - **Valid Argument:** If all premises are true, the conclusion is guaranteed to be true.
 - *Example:*
 - Premise 1: All humans are mortal.
 - Premise 2: Your seatmate is a human.
 - Conclusion: Therefore, your seatmate is mortal.
 - **Invalid Argument:** The premises could all be true, but the conclusion might still be false.
 - *Example:*
 - Premise 1: If it rains, the ground is wet.
 - Premise 2: The ground is wet.
 - Conclusion: Therefore, it is raining (Invalid, because the ground could be wet from another source like watering a garden).

Lesson 2: Language of Propositional Logic

Core Concepts

- **Propositional Logic:** The branch of logic analyzing how simple true/false statements combine using connectors (like AND, OR, NOT).
- **Single Proposition:** A standalone statement containing just one idea (e.g., "The sky is blue").
- **Compound Proposition:** Two or more statements glued together with logical operators (e.g., "The sky is blue AND it is raining").
- **Truth Table:** A cheat sheet mapping out every possible True/False combination for a set of statements.

AND, OR, NOT, XOR, CONDITIONAL, BICONDITIONAL

1. XOR (Exclusive OR: $P \oplus Q$)

P	Q	$P \oplus Q$
T	T	F
T	F	T
F	T	T
F	F	F

2. Biconditional ($P \leftrightarrow Q$)

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

3. Conditional ($P \rightarrow Q$)

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

P	Q	R	$\neg Q$	$P \wedge \neg Q$	$R \oplus Q$	$Q \vee R$	$(P \wedge \neg Q) \vee (R \oplus Q)$	$[P \rightarrow (Q \vee R)]$	$[(P \wedge \neg Q) \vee (R \oplus Q)] \leftrightarrow [P \rightarrow (Q \vee R)]$
T	T	T	F	F	F	T	F	T	F
T	T	F	F	F	T	T	F	T	T
T	F	T	T	T	T	T	T	T	T
T	F	F	T	T	F	F	T	T	F
F	T	T	F	F	F	T	F	F	F
F	T	F	F	F	T	T	F	T	T
F	F	T	T	F	T	T	T	T	T
F	F	F	T	F	F	F	F	T	F

XOR: If parehas ang value, automatic false

Biconditional: If hindi parehas ang value, automatic false

Conditional: False lang kung "IF" is T and "THEN" is F